

Week 1 : Algorithms Data Structures

Exercise 2: E-commerce Platform Search Function

```
import java.util.Arrays;
import java.util.Scanner;

// Product class
class Product {
    int productId;
    String productName;
    String category;

    public Product(int productId, String productName, String category) {
        this.productId = productId;
        this.productName = productName;
        this.category = category;
    }

    public int getProductId() {
        return productId;
    }

    public String getProductName() {
        return productName;
    }

    public String getCategory() {
        return category;
    }

    @Override
    public String toString() {
```

```

        return "Product{" +
            "productId=" + productId +
            ", productName=\"" + productName + "\" +
            ", category=\"" + category + "\" +
            "}";
    }
}

```

```

public class EcommercePlatform {

```

```

    // Linear Search

```

```

    public static int linearSearch(Product[] products, String productName) {
        for (int i = 0; i < products.length; i++) {
            if (products[i].getProductName().equalsIgnoreCase(productName)) {
                return i;
            }
        }
        return -1;
    }

```

```

    // Binary Search

```

```

    public static int binarySearch(Product[] products, String productName) {
        int low = 0, high = products.length - 1;

        while (low <= high) {
            int mid = low + (high - low) / 2;
            int comparison = products[mid].getProductName()
                .compareToIgnoreCase(productName);

            if (comparison == 0)
                return mid;
        }
    }

```

```

        else if (comparison < 0)

            low = mid + 1;

        else

            high = mid - 1;

    }

    return -1;

}

```

```

public static void main(String[] args) {

    Scanner scanner = new Scanner(System.in);

    // Array for Linear Search (unsorted)
    Product[] linearProducts = {

        new Product(1, "Laptop", "Electronics"),
        new Product(2, "Smartphone", "Electronics"),
        new Product(3, "Tablet", "Electronics"),
        new Product(4, "Headphones", "Accessories"),
        new Product(5, "Smartwatch", "Accessories")

    };

    // Sorted array for Binary Search
    Product[] binaryProducts = linearProducts.clone();
    Arrays.sort(binaryProducts,

        (p1, p2) -> p1.getProductName().compareToIgnoreCase(p2.getProductName()));

    System.out.print("Enter the product name to search: ");

    String productName = scanner.nextLine();

    // Linear Search
    int linearIndex = linearSearch(linearProducts, productName);

    if (linearIndex != -1)

```

```

        System.out.println("Linear Search: Found " + linearProducts[linearIndex]);
    else

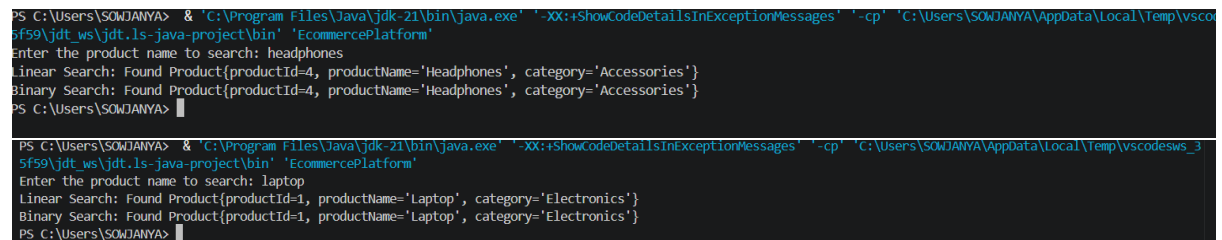
        System.out.println("Linear Search: Product not found.");

    // Binary Search
    int binaryIndex = binarySearch(binaryProducts, productName);
    if (binaryIndex != -1)
        System.out.println("Binary Search: Found " + binaryProducts[binaryIndex]);
    else
        System.out.println("Binary Search: Product not found.");

    scanner.close();
}
}

```

Output:



```

PS C:\Users\SOWJANYA> & "C:\Program Files\Java\jdk-21\bin\java.exe" "-XX:+ShowCodeDetailsInExceptionMessages" "-cp" "C:\Users\SOWJANYA\AppData\Local\Temp\vscod
5f59\jdt_ws\jdt.ls-java-project\bin" "EcommercePlatform"
Enter the product name to search: headphones
Linear Search: Found Product{productId=4, productName='Headphones', category='Accessories'}
Binary Search: Found Product{productId=4, productName='Headphones', category='Accessories'}
PS C:\Users\SOWJANYA>

PS C:\Users\SOWJANYA> & "C:\Program Files\Java\jdk-21\bin\java.exe" "-XX:+ShowCodeDetailsInExceptionMessages" "-cp" "C:\Users\SOWJANYA\AppData\Local\Temp\vscodesws_3
5f59\jdt_ws\jdt.ls-java-project\bin" "EcommercePlatform"
Enter the product name to search: laptop
Linear Search: Found Product{productId=1, productName='Laptop', category='Electronics'}
Binary Search: Found Product{productId=1, productName='Laptop', category='Electronics'}
PS C:\Users\SOWJANYA>

```

Exercise 7: Financial Forecasting

```

public class FinancialForecast {

    // Recursive method to calculate future value
    public static double futureValue(double principal, double rate, int years) {
        if (years == 0) {
            return principal; // Base case
        }
        return futureValue(principal, rate, years - 1) * (1 + rate);
    }
}

```

```

public static void main(String[] args) {

    double principal = 10000.0; // Initial investment

    double annualRate = 0.05; // 5% annual growth

    int years = 5;

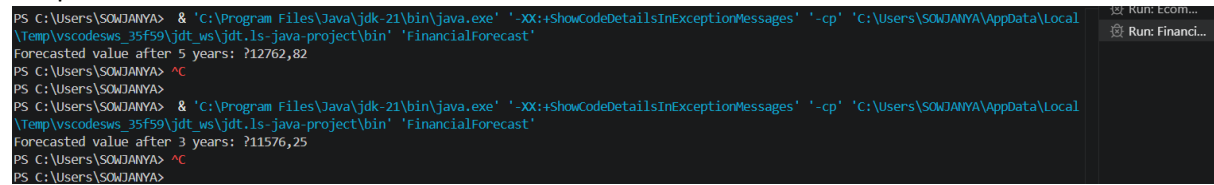
    double forecastedValue = futureValue(principal, annualRate, years);

    System.out.printf("Forecasted value after %d years: ₹%.2f\n", years, forecastedValue);

}
}

```

Output:



```

PS C:\Users\SOMDANIYA> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\SOMDANIYA\AppData\Local\Temp\vscodesws_35f59\jdt_ws\jdt.ls-java-project\bin' 'FinancialForecast'
Forecasted value after 5 years: ₹12762,82
PS C:\Users\SOMDANIYA> ^C
PS C:\Users\SOMDANIYA> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\SOMDANIYA\AppData\Local\Temp\vscodesws_35f59\jdt_ws\jdt.ls-java-project\bin' 'FinancialForecast'
Forecasted value after 3 years: ₹11576,25
PS C:\Users\SOMDANIYA> ^C
PS C:\Users\SOMDANIYA>

```